Agent Search Model

Inputs: Study Properties

- **Study properties**
  - N: number of units to check for faults
  - `resources`: total available search budget
  - `benefit`: utility of discovering one fault
  - `cost`: cost of one additional search step
  - Randomly generated study data:
    - `true_effect`: true effect size θ
    - `true_K`: true number of faults K
    - `fault_indicators`: vector, whether each item is faulty
    - `error_sizes`: error contribution of each fault

Inputs: Agent Properties

- Prior on total faults:
  - `sbj_prob_alpha` $= \alpha_K$
  - `sbj_prob_beta` $= \beta_K$
  - $K \sim \mathrm{BetaBinomial}(N, \alpha_K, \beta_K)$
- Prior on true effect:
  - `sbj_effect_mu` $= \mu_\theta$
  - `sbj_effect_sigma2` $= \sigma_\theta^2$
  - $\theta \sim \mathcal{N}(\mu_\theta, \sigma_\theta^2)$
- Prior on fault sizes:
  - `sbj_error_mu` $= \mu_e$
  - `sbj_error_kappa` $= \kappa_0$
  - `sbj_error_var_alpha` $= \alpha_e$
  - `sbj_error_var_beta` $= \beta_e$

$$e_j \sim \mathcal{N}(\mu, \sigma^2)$$

$$\mu \mid \sigma^2 \sim \mathcal{N}\left(\mu_e, \frac{\sigma^2}{\kappa_0}\right), \qquad \sigma^2 \sim \mathrm{InvGamma}(\alpha_e, \beta_e)$$

Outputs

- `stopped_in_round`
- `stopping_reason`
- `final_resources`
- `n_faults_total` $= K$
- `n_faults_discovered` $= b_{\text{final}}$
- `true_effect` $= \theta$
- `init_obs_effect` $= y_0$
- `final_obs_effect` $= y_{\text{final}}$

Observed Effect: Initial and Current

- Initial observed effect:

$$y_0 = \theta + \sum_{j=1}^{K} e_j$$

- After round i, with discovered error sizes $x_1, \ldots, x_{b_i}$:

$$y_i = y_0 - \sum_{j=1}^{b_i} x_j$$

- State variables:
  - i: current round
  - b_i: faults discovered by round i
  - y_i: remaining observed effect after corrections

Agent Decision Making

1. The agent forms a posterior $\pi_i(k)$ over total faults K.
2. With this, we calculate the number of expected faults remaining.
3. From this, the agent derives a belief p_{i+1} about the probability that the next check finds a fault.
4. This is compared to costs and benefits as a expected utility criterion.
5. The agent decides to continue or stop based on this criterion and whether the remaining resources would be sufficient for another round of searching.

Posterior over Total Faults K

The posterior is based on three likelihood components:

1. Fault-count likelihood: $P(b_i \mid K = k, i, N)$
2. Fault-size likelihood: $P(x_1, \dots, x_{b_i})$
3. Currently remaining effect likelihood: $P(y_i \mid K = k)$

Initial prior:

$$K \sim \mathrm{BetaBinomial}(N, \alpha_K, \beta_K)$$

Unnormalized posterior:

$$\tilde{\pi}_i(k) = P(K = k)\, P(b_i \mid K = k, i, N)\, P(x_1, \dots, x_{b_i})\, P(y_i \mid K = k)$$

Normalized posterior:

$$\pi_i(k) = \frac{\tilde{\pi}_i(k)}{\sum_{m=0}^{N} \tilde{\pi}_i(m)}$$

Posterior: Fault-Count Likelihood

If b_i faults have been found in i inspected units, then for candidate total faults k:

$$P(b_i \mid K = k, i, N) = \frac{\binom{k}{b_i}\binom{N-k}{i-b_i}}{\binom{N}{i}}$$

- Hypergeometric likelihood

Posterior: Likelihood of Observed Fault Sizes

For observed fault sizes $x_1, \ldots, x_{b_i}$:

$$P(x_1, \ldots, x_{b_i}) = \iint \left[\prod_{j=1}^{b_i} \mathcal{N}(x_j \mid \mu, \sigma^2) \right] p(\mu, \sigma^2) \, d\mu \, d\sigma^2$$

- Marginal likelihood under the Normal–Inverse-Gamma prior
- Uses conjugate updating of (μ, σ^2)
- Depends on the observed fault sizes, not directly on k

Posterior: Likelihood of Remaining Observed Effect

Then the remaining observed effect under the number of undiscovered faults $r_i(k) = k - b_i$ is modeled as

$$y_i \mid K = k \sim \mathcal{N}\big(\mu_\theta + r_i(k)\mu_e,\ \sigma_\theta^2 + r_i(k)\sigma_e^{2*}\big)$$

with plug-in error variance

$$\sigma_e^{2*} = \frac{\beta_e}{\alpha_e - 1}$$

So

$$P(y_i \mid K = k) = \phi\big(y_i; \mu_\theta + r_i(k)\mu_e,\ \sigma_\theta^2 + r_i(k)\sigma_e^{2*}\big)$$